

# Entity Relationship Model

## Topic 4

# LOs

- By the end of this topic you should be able to:
  - 1. Determine if a particular factor is an entity or a non entity;
  - 2. Give examples of five types of attributes that exist;
  - 3. Define the terminologies that exist in an Entity Relationship Model; and
  - 4. Construct an Entity Relationship Diagram (E-R Diagram).

# Introduction to Modelling

- Model can make it easy to understand and communicate by giving pictures to the text or information to be presented.
- Because of this, the Entity-Relationship Model is used by database designers and end users to analyse data.
- Why? Because the E-R Model contains symbols which represent data structure.
- It is able to explain types of data, characteristics of data and relationship between data.
- Each of these symbols is named entity, attribute and relationship.
- Although this model is made up of a combination of only three symbols, this is enough to depict the data structure required by the organisation.

# ERD

- The E-R Model is used to construct the conceptual data model. It represents the structure and constraints of a database which is not dependent on any software (like Database Management System) or any data model to be used to implement a database.

# 1- ENTITY

- Entity is an object of the real world which can store data and can be defined with clarity.
- Object may consist of something that exists physically such as
  - individual,
  - place,
  - article or
  - something that exists conceptually such as an event or concept in a user environment where data is to be stored.

# 1- ENTITY

- As such, when you come across an object, ask yourself:
  - Do you wish to store data related to that particular object?
  - Is it capable of storing data?
  - Is it related to or important in the application to be constructed.

# 1- ENTITY

| ■ Entity      | Examples                      |
|---------------|-------------------------------|
| ■ Individual  | Customer, Worker, Supplier    |
| ■ Venue/place | District, State, Country      |
| ■ Object      | Stationary, Furniture         |
| ■ Event       | Purchase Order, Booking, Sale |
| ■ Concept     | Project, Account, Unit        |

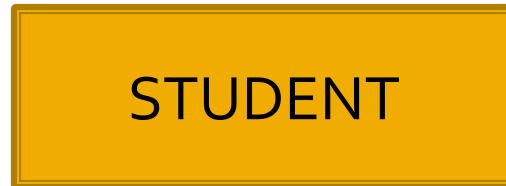
Table 4.1: Entities in the Environment of a Firm

# 1- ENTITY

- an environment cannot become an entity under any circumstance because environment is expressed by other entities that exist inside it, and not expressed by the environment itself.
- Entities consist of entity name or entity type.
- Entities also consist of entity occurrence i.e. objects modeled as entities.
- For example, WORKER consists of Abu, Ramli and Chin which represents the entity occurrence for the worker type of entity.

# 1- ENTITY

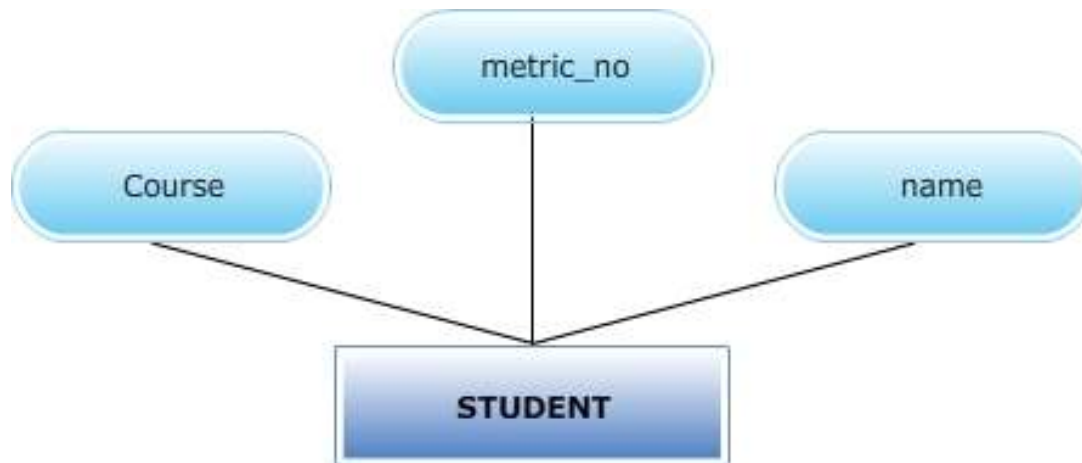
- Entity is represented by the symbol of a rectangle and labeled with a noun in the middle of the symbol. The following is an example of an entity:



## 2- ATTRIBUTES

- A description or an entity characteristic is known as an attribute
- As a student, what differentiates you from other students? You are differentiated through metric number, faculty, course and address.
- These are attributes of you as a student. For a worker, attributes consist of name, worker number, department and address.

## 2- ATTRIBUTES



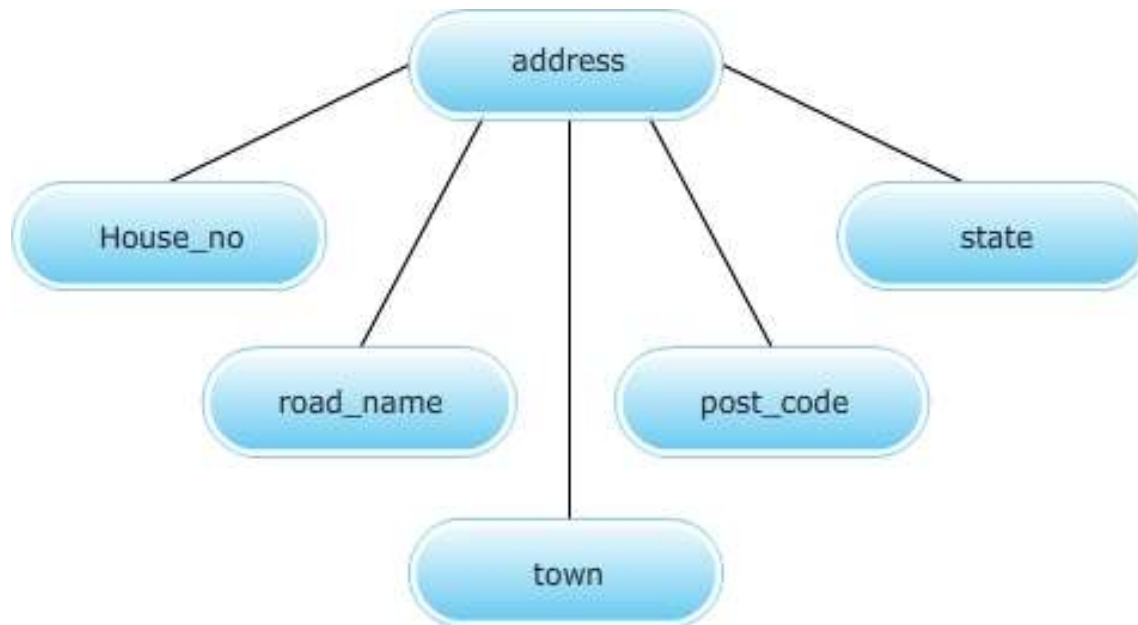
# 2-Type of Attribute

- Simple Attributes vs. Composite Attributes
- A simple attribute
  - has only one component,
  - exists independently
  - and cannot be broken up.
  - Independence means that the component does not have to rely on other attributes.
  - For example, a name or gender - both cannot be divided into two or more sections.

# 2-Type of Attribute

- Simple Attributes vs. Composite Attributes
- Composite attribute
  - comprises of many components, each one existing independently.
  - An example is an **address** with sub-attributes such as house number, road name, postcode, district, state and country.

## 2- ENTITY-Composite attribute



# 2-Type of Attribute

- Solitary Value Attribute vs. Multiple Value Attribute
- Solitary value attribute
  - an attribute that consists of only a single value.
  - Metric number and identity card number are examples of solitary attributes.
  - A single attribute is not necessarily a simple attribute. For example, the serial number of a product is MA-22-030800.
  - This is a solitary attribute but it is also a composite attribute because it can be divided into MA-factory, 22 shifts in the factory and 030800-product number.
- A multiple value attribute
  - an attribute consisting of many values.
  - For example, an individual can have several telephone numbers such as house telephone number, office telephone number and mobile telephone number

## 2-Type of Attribute

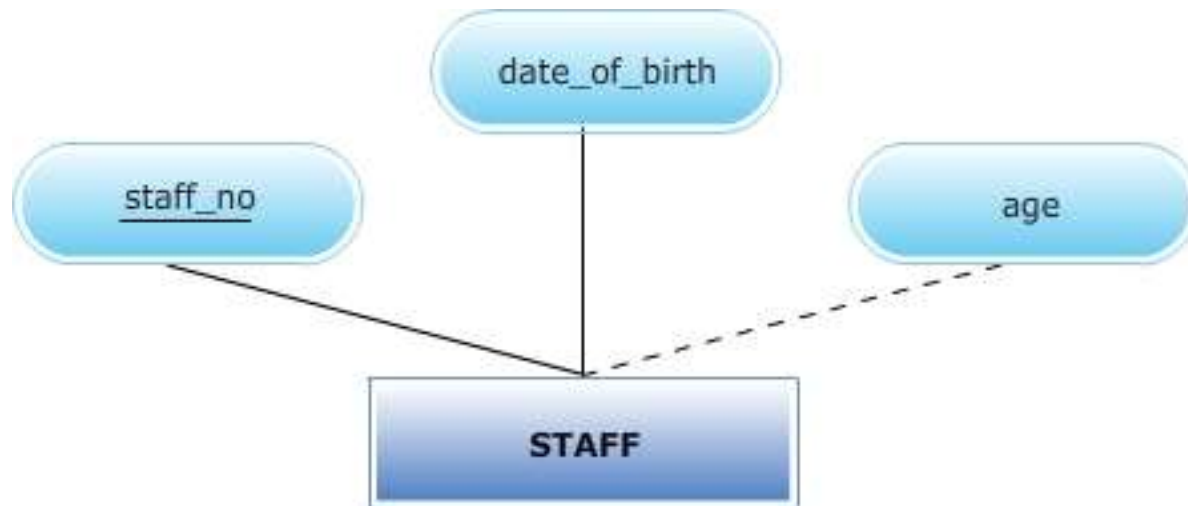
- Solitary Value Attribute vs. Multiple Value Attribute
- A multiple value attribute
  - an attribute consisting of many values.
  - For example, an individual can have several telephone numbers such as house telephone number, office telephone number and mobile telephone number

# 2-Type of Attribute

## ■ Derived Attribute

- A derived attribute is an attribute where its **value is derived from the value of related attributes or set of other attributes.**
- An example is the age attribute which is derived from the date of birth. This derived attribute is represented by the oval symbol, the same as for other attributes, but it is connected to the entity by **broken lines**

## 2-Type of Attribute



## 2- Attribute Domain

- **Attribute Domain**
- Attribute domain is a set of values for an attribute. For example, the attribute domain for staff number is integer (i-30). The types of integers are:
  - (i) Character
  - (ii) Numeric
  - (iii) Date

## 2- Attribute Domain

- Integers in brackets refer to integer type as well as the highest and lowest value.
- As an example, the domain attribute for staff number is integer (1 - 30), then we understand that the domain consists of numeric integer with 1 as the lowest value and 30 as the highest value.

## 2- Attribute Null Value

- **Null Value**
- Attribute value which does not exist, unknown at certain times or not related. In this kind of situation the value of the attribute is empty (not filled with any value).
- Does not mean that the value of the attribute is zero (0). For example, if you do not take part in a quiz for no particular reason then your marks will be 0 (quiz value) but if you do not take part in the quiz due to illness, your quiz value is empty (null value) until such time as you can take the quiz stated.
- Some attributes are given the null value because the value bears no relation to the entity occurrence. An example is the office address attribute being given the null value for those unemployed.

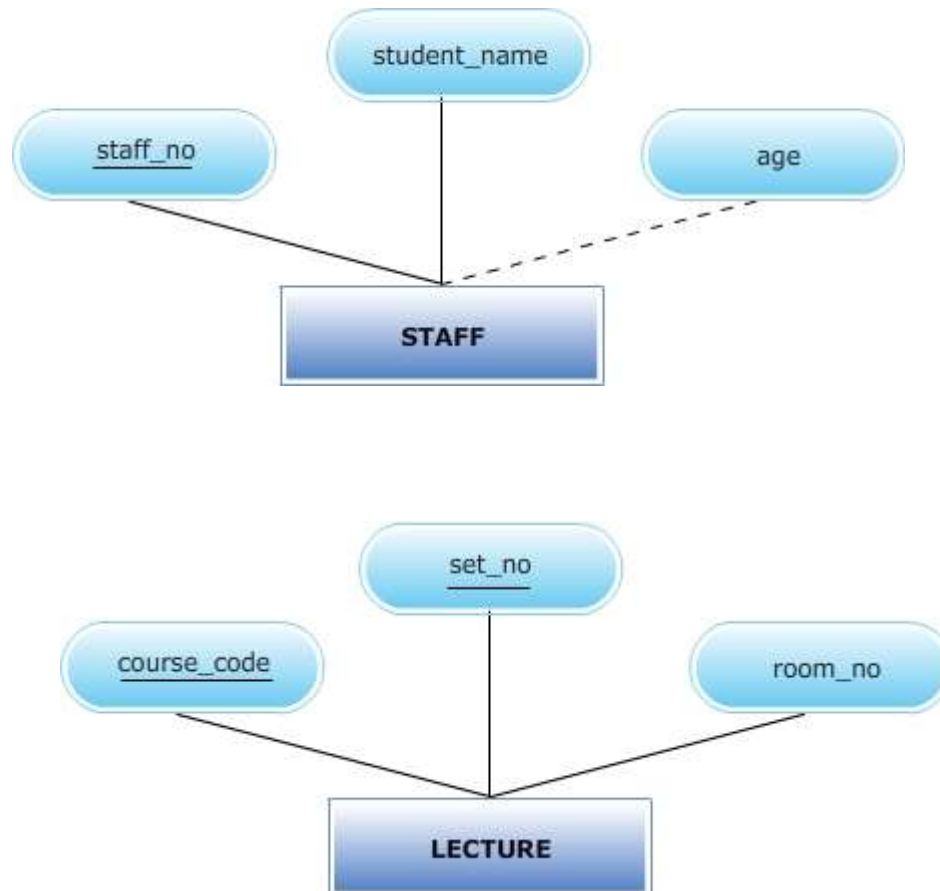
# 3-KEYS

- **KEYS**
- Each entity usually has attributes with different values. Each entity occurrence for STUDENT has a unique metric number to differentiate each student.
- Even if there are many other students with the same name as yours, you can still be easily identified through your metric number. This is because no two students will have the same metric number.
- This attribute is called key.

# 3-KEYS

- The key is
  - one or several attributes which can differentiate the entities they describe.
  - The attribute value of these keys must be unique and
  - must not have the null value.
  - Key attributes are underlined in the E-R Model to differentiate them from other attributes which are not keys.

# 3-KEYS



# 4- RELATIONSHIP

- A relationship is the link between entities. The relationship symbol is represented by a diamond shape, whereas the label for relationship is a verb or conjunction. It links the two entities by a line.

# 4- RELATIONSHIP



- LECTURER is linked to COURSE through the relationship teach as shown in Figure 4.8.
- Other examples are CUSTOMER makes RESERVATION and PILOT flies AIRPLANE. Words in italics represent the relationship between one entity to another

# 4- RELATIONSHIP

- **Type of Relationship**
- You have learned that entities can be connected to each other through relationship.
- Actual relationships can involve up to three entities.
- The number of entities involved is referred to as degree of relationship. There are three degrees of relationship.

# 4- RELATIONSHIP

- Unary Relationship (Recursive)
- Unary relationship, also called recursive relationship, is a relationship involving only one entity.
- Relationship is established between occurrences that happen within the same entity.
- For example, STAFF manages STAFF (Professor Yusof, dean, manages Pn. Sarah, lecturer: both are occurrences of the staff entity.
- This example is depicted in Figure 4.9a). Similarly, the entity COURSE is a prerequisite of COURSE (IT introductory course is a prerequisite for course in analysis and system design).

# 4- RELATIONSHIP



# 4- RELATIONSHIP

- Binary Relationship
- Binary Relationship is a relationship between two entities. It is the most common relationship used in any data model.
- For example STUDENT registers for a COURSE (depicted in Figure 4.b); PATIENT takes MEDICINE, CUSTOMER buys PRODUCT.

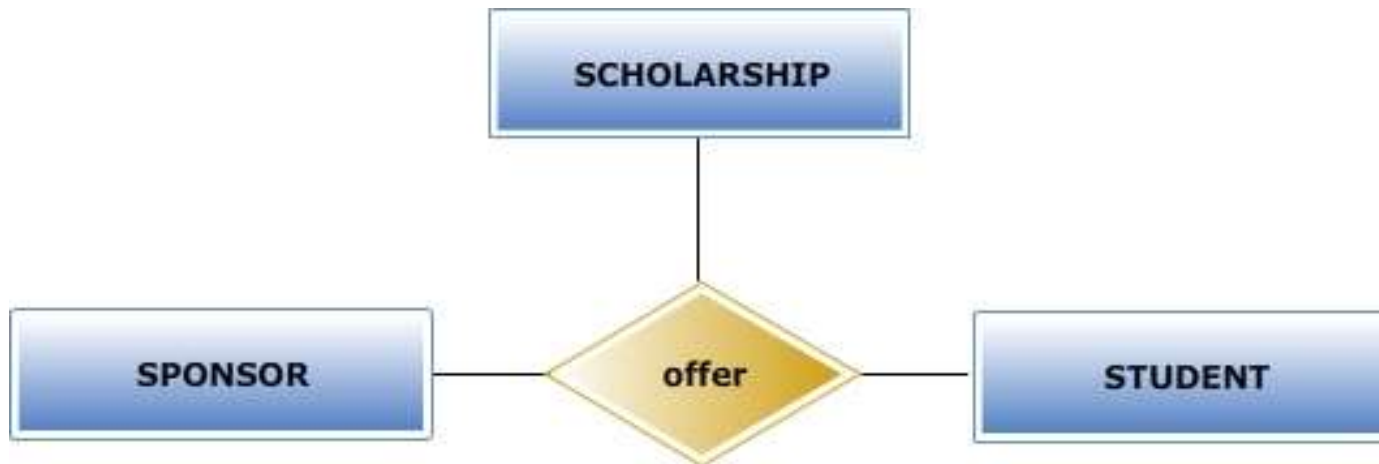
# 4- RELATIONSHIP



## 4- RELATIONSHIP

- Ternary Relationship
- Ternary relationship is a simultaneous relationship between three entities. For example, SPONSOR offers SCHOLARSHIP to STUDENT.
- The SCHOLARSHIP offered to STUDENT involves both the SPONSOR and the SCHOLARSHIP.
- We cannot offer the SCHOLARSHIP without involving all the three entities. As such, this relationship constitutes ternary relationship, the example of which is shown in Figure 4.9c as follows:

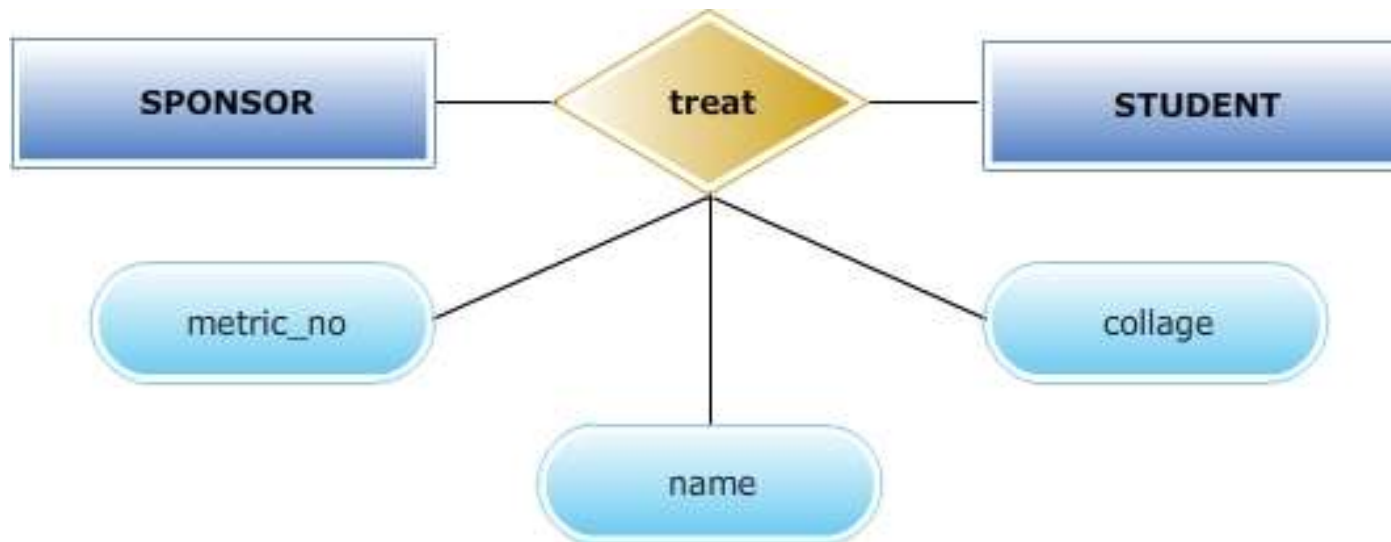
# 4- RELATIONSHIP



# Relationship Attribute

- As with other entities, relationship can contain attributes.
- Like entity attributes which describe the entity, the relationship attribute describes the relationship.
- An example is the treatment relationship (between DOCTOR and PATIENT) which consists of type of treatment, type of ailment and medicine as shown in Figure 4.9d.
- Normally, the existence of relationship attribute indicates the probability of the existence of entities which were not identified during data analysis.
- If the treatment relationship has many relationship attributes, it is more suitable as an entity compared to relationship..

# Relationship Attribute



# Cardinality

- **Cardinality**
- Relationship cardinality describes the **number of relationships between one entity to other entities**. In other words, it is the **number of entity occurrences related to other entity occurrences**.
- The actual number of these entity connections depends on the policy of the organisation.
- Observe that any data model which you have encountered before this is an example which represents specific situations only, not general representation.
- For example, LECTURER in a particular university may teach only one course whereas LECTURER in another university may teach many COURSES.

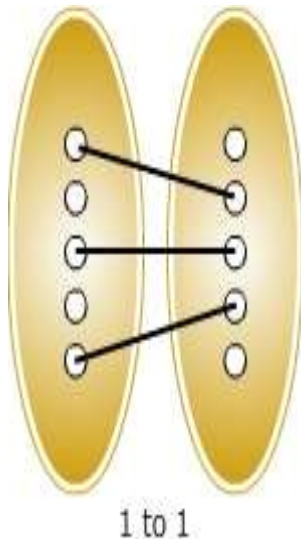
# Cardinality

- The type of relationship is determined by the organization where the database is constructed. Cardinality is determined by the policy of the organization called **business regulations**.
- There are 3 types of relationship cardinality i.e.:
  - One-to-one relationship;
  - One-to-many relationship; and
  - Many-to-many relationship.

# Cardinality

## ■ 1:1 Relationship

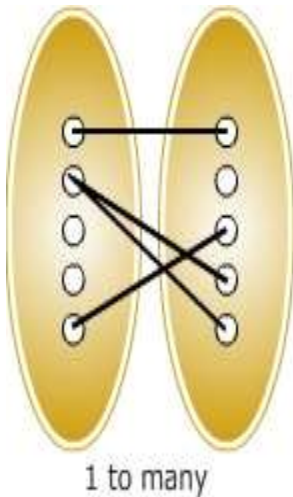
- One occurrence in entity A is associated to **one** occurrence in entity B and **one** occurrence in entity B is associated to one occurrence in entity A.



# Cardinality

## ■ 1:M Relationship

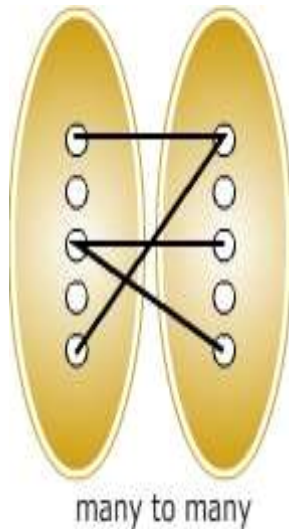
- One occurrence in entity A is associated to **many** occurrences in entity B and one occurrence in entity B is associated to one occurrence in entity A.



# Cardinality

## ■ M:N Relationship

- One occurrence in entity A is associated to **many** occurrences in entity B and **one occurrence in entity B is associated to many occurrences in entity A.**



# Relationship Participation

- Entity participation in a relationship can be classified as **compulsory or optional**. Optional means an occurrence in **entity A may not be associated to entity B every time**.
- In other words, if the minimum cardinality is 0, participation is optional; if the minimum cardinality is 1, participation is compulsory. For example, in a LECTURER teach COURSE relationship, there are LECTURERS who do not teach any COURSE in a particular semester (due, maybe, to study leave, sabbatical leave or maternity leave). Therefore, the cardinality of lecturer and course is 0 and participation of LECTURER toward COURSE is optional.

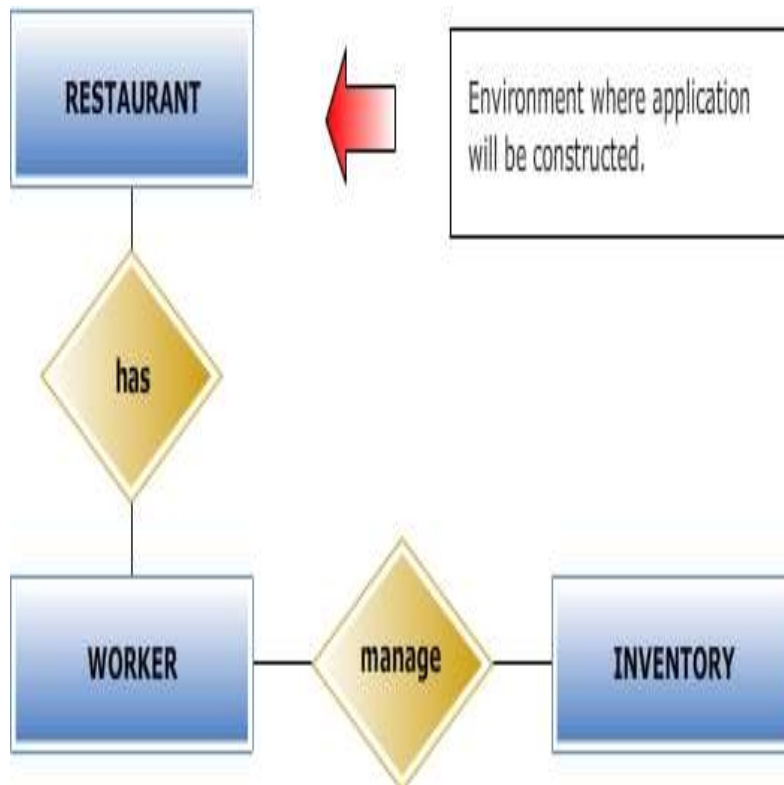


# Relationship Participation

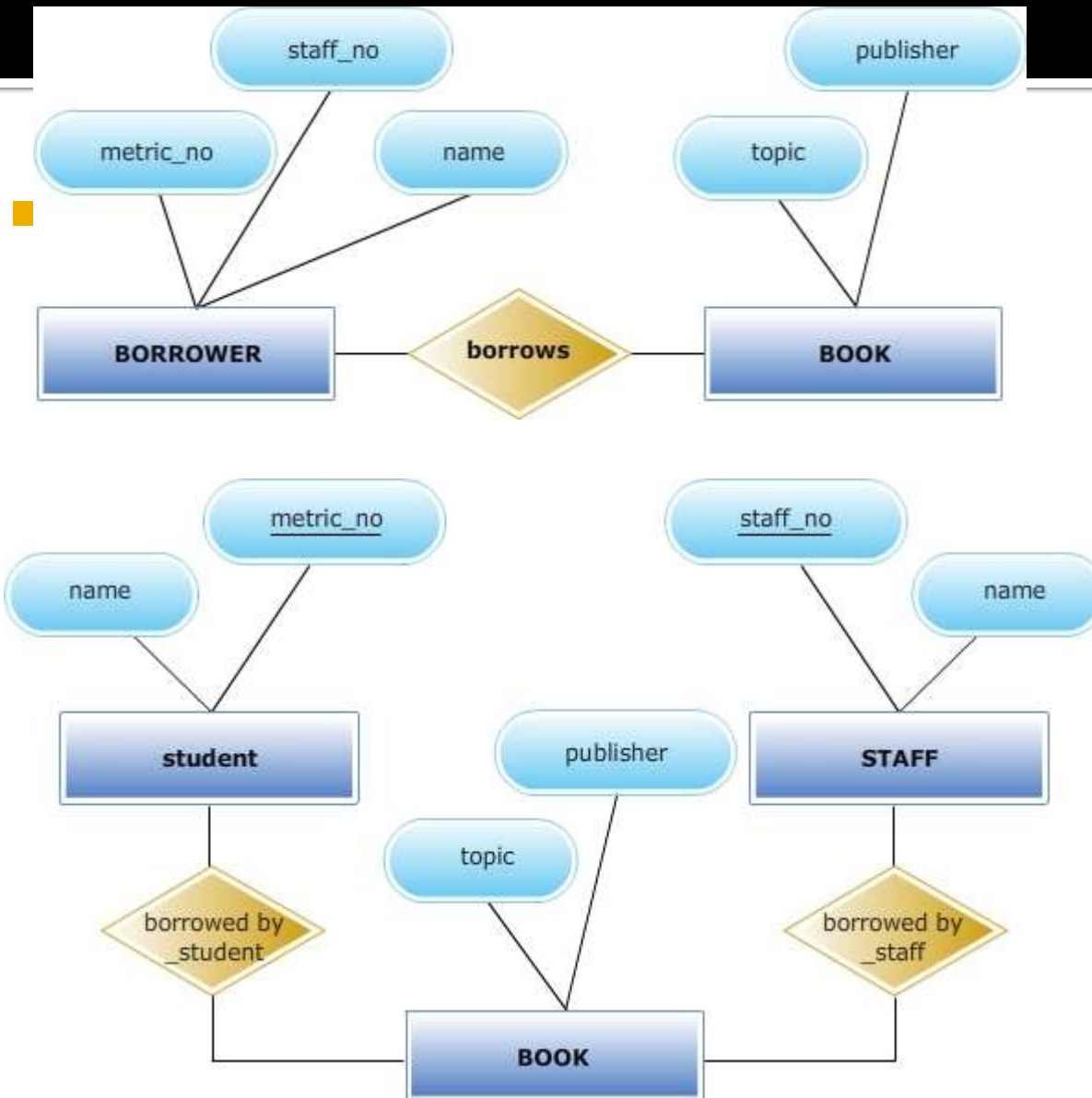
- Let us look at COURSE participation in relation to LECTURER. If a LECTURER who is supposed to teach a particular COURSE is absent, another LECTURER will be assigned to teach the COURSE.
- Each COURSE will be taught by at least one LECTURER. In this case, the minimum cardinality is 1.
- As such, COURSE participation towards lecturer is compulsory. Therefore, you can conclude that **participation is determined by a two way relationship, similar to relationship cardinality.**

# GUIDELINES ER MODEL

- Do not insert the system environment which is modeled as a component of the E-R



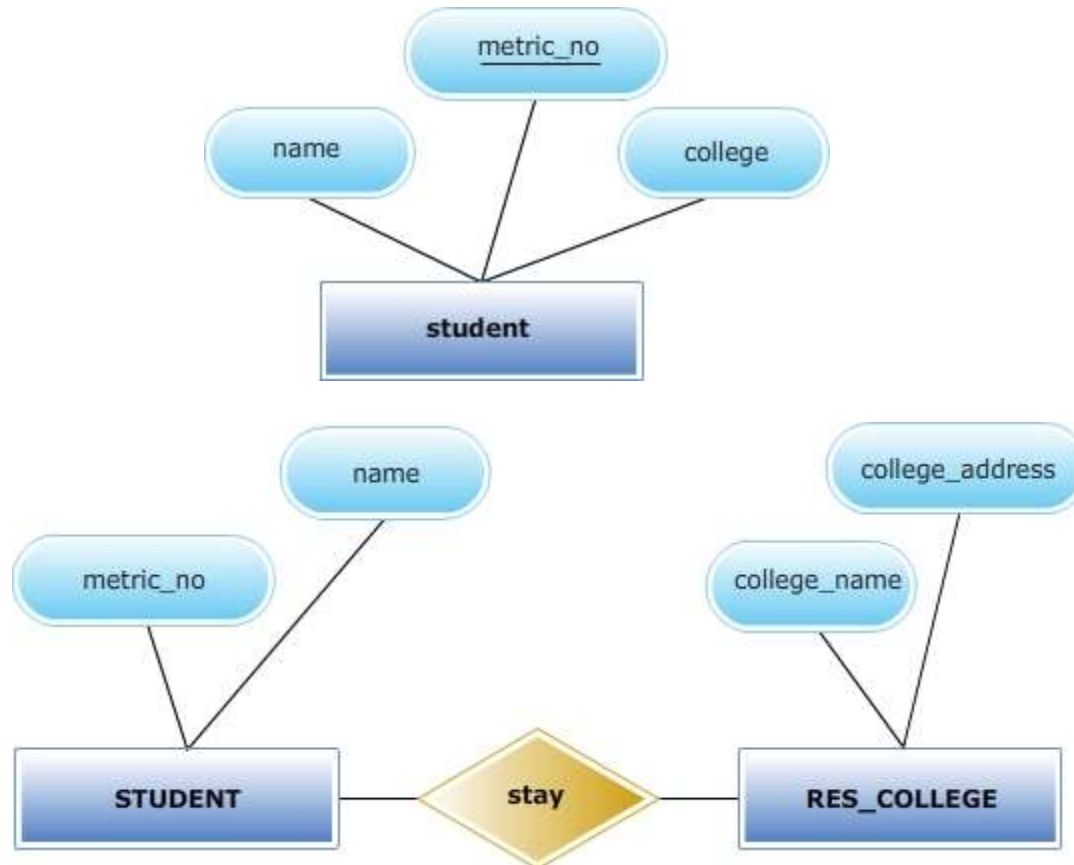
# GUIDELINES ER MODEL



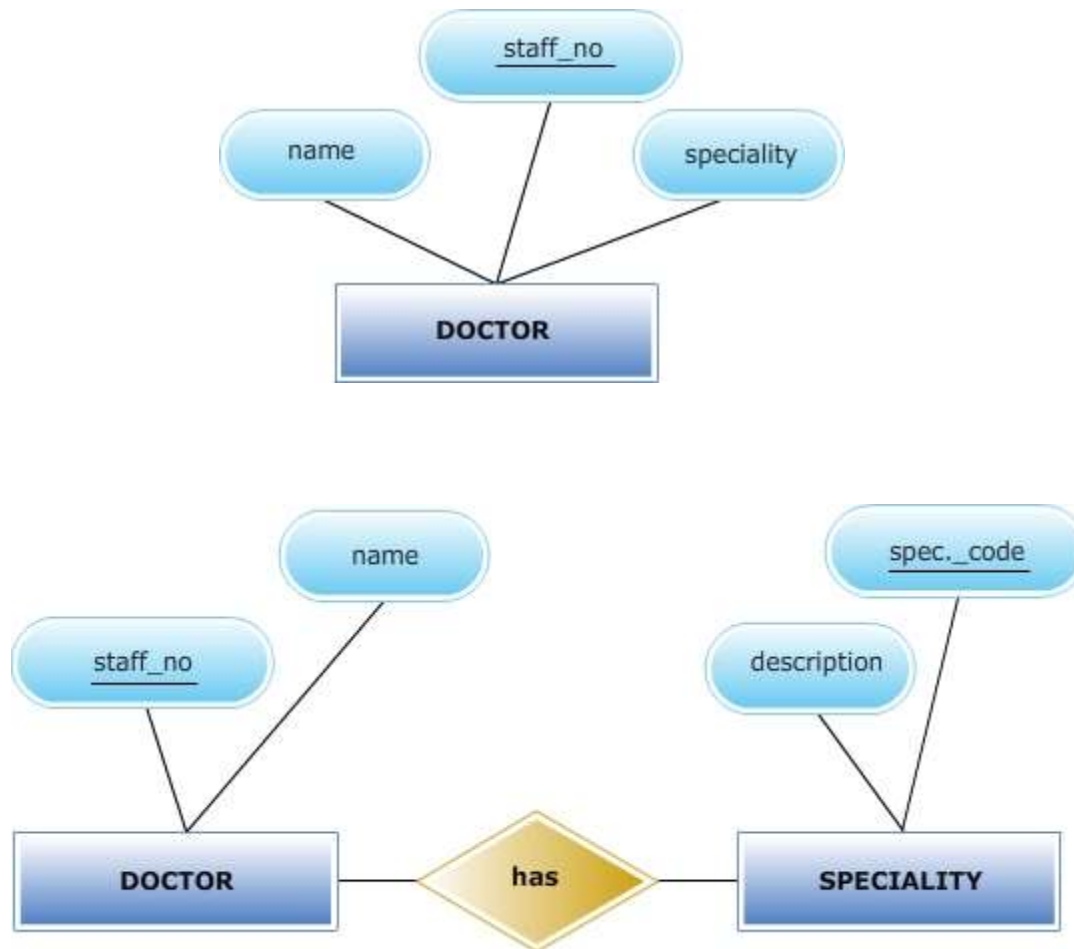
Attribute and key for a particular environment is not necessarily the same for other environments

# GUIDELINES ER MODEL

■ attribute can be made into an entity.

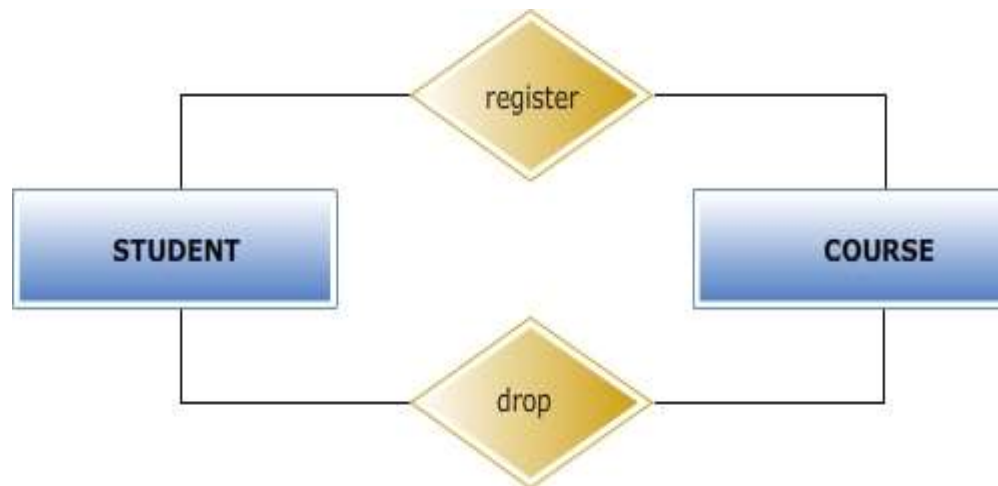


# GUIDELINES ER MODEL



Convert multiple value attributes as entities

# GUIDELINES ER MODEL



Two entities can have more than one relationship

# GUIDELINES ER MODEL

1. Determine entities and the relationship between them. Start first with the main entity followed by other entities.
2. Determine the attributes related to the entities.
3. Determine the attributes related to relationship (if there are any)
4. Choose the keys for the entities.
5. Determine the domain for each attribute.
6. Combine the diagrams of entity, relationship and attribute to develop a complete E-R model. The combination must be made without allowing entities to exist independently (hanging).
7. Thoroughly check and refine the E-R model if necessary (discuss with users).